

Chaeyoon Kim

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Certified AI Ethicist and Data Scientist with 4+ years of UK public sector experience, following 6+ years in semiconductor engineering at Samsung Electronics. Specialised in large-scale analytics, AI engineering, and healthcare workforce modeling. Proven track record of building LLM-powered applications in Python and PySpark, while leading reproducible, secure analytics engineering for the NHS workforce across England.

EXPERIENCE

Data Scientist – NHS England (London, UK)	Apr 2024 – Present
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- Redesigned the CT1 training-place allocation methodology from a weighted multi-criteria model into a rule-based model using Python and Excel, presenting Tableau outputs that informed prioritisation for under-doctored deaneries during 2024 BMA industrial action negotiations.
- Lead Developer for the 10-Year Workforce Plan modelling under NHS England's Business Critical Model (BCM) governance: validating the model's key assumptions in Python and PySpark, working with Department of Health and Social Care and NHS England policy leads to shape the response to National Audit Office recommendations, and presenting model components to the External Scrutiny Panel.
- Manage the project's dedicated Azure Data Lake Storage (ADLS Gen2) and Databricks clusters, coordinating with UDAL Data Engineers on secure environments; co-led the migration of legacy PySpark pipelines off deprecated DBFS onto Entra-secured Data Lake containers and coordinated Windows 11 migration testing to protect model continuity.
- Led the transition of legacy Excel workforce models into automated Databricks and Python pipelines with documented assumptions, strengthened data lineage and peer-reviewed Git version control; refactored the Python codebase to cut model runtime by around half and enable cross-functional teams to extend it.
- As Community Pharmacy modelling workstream lead, coordinated a dozen stakeholder groups — including pharmacy Deans and Chief Pharmaceutical Officers — into a single delivery scope; launched a Pythonised supply-and-demand model within six months, secured the model SRO's approval despite late changes to baseline and counterfactual assumptions, and presented it at the national consultation on Community Pharmacy workforce interventions.
- Designed an NLTK-based sentiment-analysis pipeline for the Reporting, Insights, Publications and Transparency programme and contributed to a paper on insights maturity for the CEO's office and Board; applied NLP and sentiment analysis to meeting transcripts to assess staff views on organisational change and internal AI adoption.
- Conduct horizon scanning of emerging data and AI engineering practice, including exploratory research into AI-enabled gains in workforce productivity.
- Providing Databricks, Python and Git training and onboarding to analysts and new data scientists, promoting coding best practices across the modelling team.

Junior Data Scientist – Health Education England (London, UK)	Jun 2022 – Mar 2024
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- Maintained a pythonised geo-spatial analysis and produced Tableau dashboards analysing clinical placement capacity, workforce supply trends and geographic demand across 65 medical specialties for the HEE Board's tariff-funding review, advising medical Deans on redistributing specialty training posts and informing NHS England's recommendations to the Office for Students on new dental school places.
- Supported R programming capability to predict a complex birth delivery on planning maternity labour services, with source data from Royal College of Obstetricians and Gynaecologists (RCOG) statisticians.
- Collaborated with Data Engineering and Product teams using Jira/Agile framework to manage project timelines.

Data Manager – UCL Great Ormond Street Institute of Child Health (London, UK)	Apr 2022 – Jun 2022
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- Collaborated with bioinformaticians and statisticians on the GSK Sotrovimab Covid-19 study, cleaning and transforming lab data.
- Designed a MySQL server for secure data transfers from entry points.

Engineer – Samsung Electronics, Foundry Business (Gyeonggi-do, South Korea)	Jul 2011 – Jan 2018
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- Operated Chemical Vapour Deposition (CVD) equipment in 24/7 shift operations for high-volume 32/28nm and 14/10nm semiconductor production, and supported set-up and ramp-up of a new 7/10nm line.

- Established maintenance SOPs and data-driven part-replacement criteria by evaluating correlations between plasma process parameters and component condition.
- Derived optimal process conditions through simulation built on deposition and clean measurements, suppressing particle generation and improving yield.
- Owned equipment asset and standards management on the Equipment Engineering System (EES): built gas and chemical usage metering and monitoring that detected unnecessary utility consumption and cut costs; with EES Master privileges, configured thin-film unit-process alarms and implemented Tool-to-Tool Matching (TTTM).

PROJECTS

Strategic planning:

- **10-Year Workforce Plan modelling:** Working with a multidisciplinary group to refresh and refine the NHS Long-Term Workforce Plan modelling (published in Jun 2022), incorporating National Audit Office feedback.

AI & Machine Learning engineering:

- **Endo Loop:** Co-built a non-diagnostic, at-home pattern journal for endometriosis and chronic pelvic pain, combining voice or manual symptom logs with wearable signals in a deterministic, safety-guarded pattern engine; awarded joint 5th place at the eMed and OpenAI Reimagine Health hackathon (Jul 2026).
- **NHS Policy Navigator:** Built an adaptive Retrieval-Augmented Generation pipeline over the published 10-Year Health Plan and public NHS news feeds at the MongoDB Agentic Evolution Hackathon (May 2026), then refined it with a working group through a spec-driven development cycle at near-zero infrastructure cost.
- **Pharmacy First agent:** Presented an LLM-powered analysis about NHS Pharmacy operation with open source at PyConUK, NHS RPySOC conference, and Health and Care Analytics (HACA) conference 2025.
- **Reducing the missed NHS appointments:** Proposed a ML solution to reduce the millions costs and increasing waitlists for a hackathon challenge set by the Number 10 Downing Street data science team (Sep 2025).
- **AI for NHS Healthcare professionals:** Co-developed a multilingual "NHS Career Coach" on Azure AI Foundry that retrieves and cites Health Education England careers content and adapts its guidance to user personas; awarded 2nd place at the NHS England, Microsoft and Kainos Hack for Health hackathon (Nov 2024).
- **Integrated Operational Pressures Escalation Level (OPEL) prediction:** Demonstrated ML models on NHS Federated Data Platform during a hackathon operated by Palantir Technologies and KPMG (Nov 2024).

Teaching and outreach:

- **AI Engineering training programme:** Designed and delivered a training programme for learners from undergraduates to NHS analysts and clinicians, teaching reproducible software engineering, responsible AI and governance principles using open-source NHS datasets; accepted as a peer-reviewed oral presentation in computer science education.

Research and publication:

- Developed R and Python conversion code for a textbook, <Foundations of Programming, Statistics, and Machine Learning for Business Analytics> (published in Apr 2023), then co-authored a translated edition in Korean (published in Feb 2025).
- Kim, C. and Jiménez-Ruiz, E. (2022) [CitySAT: a System for the Semantic Answer Type Prediction Task](#). CEUR Workshop Proceedings, Vol. 3119 (SMART 2021, ISWC Semantic Web Challenge), pp. 77–88. First author.

EDUCATION

MSc Data Science (Distinction) – City St George's, University of London (London, UK)

Sep 2020 – Oct 2021

- Ranked 1st on the global leaderboard of the International Semantic Web Conference (ISWC) 2021 SMART Challenge as a sole entrant with CitySAT, a two-stage hybrid system using Logistic Regression for answer category and literal types and a Multi-Layer Perceptron over ~760 DBpedia ontology classes; achieved 98.4% accuracy and 84.2% NDCG@5 on the official test set, and delivered an oral presentation and a poster at ISWC in October 2021.

PgCert Data Analytics – University of Sheffield (Sheffield, UK)

Sep 2018 – Jul 2020

- **Modules include** Computer Security and Forensics, Machine Learning and Adaptive Intelligence, Statistical Data Science in R.

BSc Electronic Engineering – Kyungpook National University (Daegu, South Korea)

Mar 2007 – Aug 2011

- **Modules include** Electronic Engineering Lab (MATLAB), Calculus, Numerical Analysis.

CERTIFICATIONS

Certified AI Ethicist (CAIE) – oxethica

Apr 2026

Oxford AI Ethics, Regulation and Compliance Programme – Saïd Business School, University of Oxford

Mar 2026

- Completed the programme ([credential](#)); published the reflective article "[Stepping back from my technical day-to-day and into history, philosophy and regulation](#)" on the Saïd Business School Oxford Experience blog.

VOLUNTEERING

The UK-Korea Global Health Forum working group: organising a conference with senior researchers at the London School of Hygiene & Tropical Medicine (LSHTM).

Professional Mentor at the University of Greenwich, and City St George's, University of London: supporting early career interview preparation for 3+ years; recognised at City St George's 2025 Professional Mentoring Awards and featured in the alumni article "[A great mentor is both generous and curious](#)".

Global Ambassador at LangChain: hosting technical meetups and organising a hackathon in central London; collaborated with AI builder communities across North and South America (including Halifax and Buenos Aires) in 2025, and with engineering communities across Europe (including Amsterdam, Munich, Zurich, Stockholm and Paris) in 2026.

Conference and community organiser: volunteer at PyCon UK and other leading UK technology conferences; planned or helped run local meetups ranging from 50 to 600+ attendees.